

MongoDB TPS, CPU & Latency Monitoring with Node Exporter + Grafana

Author: Venkata Subrahmanyeswarao (implementation)

Helper: ChatGPT

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Overview

This document describes how to implement end-to-end monitoring of MongoDB transactions per second (TPS), CPU utilization, and latency using:

- A custom Bash script that:
 - Reads TPS metrics from ``mongostat``
 - Reads latency metrics from ``db.serverStatus().opLatencies``
 - Reads MongoDB process CPU from `/proc`
 - Exposes everything as Prometheus metrics via Node Exporter textfile collector
- Node Exporter (with textfile collector)
- Prometheus
- Grafana dashboards and panels

The steps are written for:

- OS: Ubuntu (MongoDB running locally)
- MongoDB: runs as ``mongod`` process
- Metrics are exposed via Node Exporter on port 9200

You can adapt hostnames, paths, and credentials as needed.

1. Prerequisites

1. A MongoDB replica set or standalone instance running.
2. Shell access (root or sudo) to the MongoDB server.
3. Node Exporter installed and running on the MongoDB node.
4. Prometheus configured to scrape Node Exporter on ``http://<server>:9200/metrics``.
5. Grafana installed and able to reach Prometheus.

You also need a MongoDB user with permission to run ``serverStatus`` and connect via TLS:

Example user (already existing in your setup):

- User: `sadmin`
- Pass: `adm!n@1234`
- Auth DB: `admin`
- Host: `192.168.61.39`

TLS files (in your server):

- CA: `/etc/mongo/ssl/ca.crt`
- PEM: `/etc/mongo/ssl/toucanint-full.pem`

2. Final Monitoring Script

Create the script directory (if not already):

```
sudo mkdir -p /mongo/scripts
cd /mongo/scripts
```

Create the script file:

```
sudo nano /mongo/scripts/mongo_cpu_tps_latency_exporter.sh
```

Paste the following FINAL script (adjust credentials if needed):

```
-----  
SCRIPT: /mongo/scripts/mongo_cpu_tps_latency_exporter.sh  
-----
```

```
#!/bin/bash  
# =====  
# MongoDB TPS + CPU (Delta) + REAL Latency Exporter (HARDENED)  
# Output → Node Exporter Textfile Collector (Grafana Ready)  
# =====  
  
OUT_FILE="/var/lib/node_exporter/textfile_collector/mongo_ops.prom"  
INTERVAL=10  
  
# ---- Credentials ----  
MONGO_USER="sadmin"  
MONGO_PASS='adm!n@1234'  
MONGO_HOST="192.168.61.39"  
AUTH_DB="admin"  
  
# ---- TLS ----  
SSL_CA="/etc/mongo/ssl/ca.crt"  
SSL_PEM="/etc/mongo/ssl/toucanint-full.pem"  
  
echo "Collecting MongoDB TPS + CPU (interval delta) + real latency..."  
echo "Output: $OUT_FILE"  
echo "Press Ctrl+C to stop."  
  
# ----- MONGODB PID -----  
MONGOD_PID=$(pgrep -x mongod | head -n 1)  
if [ -z "$MONGOD_PID" ]; then  
    echo "■ mongod process not found"  
    exit 1  
fi  
  
# ----- INITIAL CPU -----  
prev_total=$(awk '/^cpu / {print $2+$3+$4+$5+$6+$7+$8}' /proc/stat)  
prev_proc=$(awk -v pid=$MONGOD_PID '{print $14+$15}' /proc/$MONGOD_PID/stat)  
  
# ----- INITIAL LATENCY -----  
read prev_lat prev_ops <<< "$(  
mongosh "mongodb://$MONGO_USER:$MONGO_PASS@$MONGO_HOST/$AUTH_DB" --tls --tlsCAFile "$SSL_CA" -  
s=db.getSiblingDB("admin").serverStatus().opLatencies;  
lat=s.reads.latency+s.writes.latency+s.commands.latency;  
ops=s.reads.ops+s.writes.ops+s.commands.ops;  
print(lat);  
print(ops);  
' 2>/dev/null | tr -cd '0-9'  
' | head -n 2  
)"  
  
prev_lat=${prev_lat:-0}  
prev_ops=${prev_ops:-0}  
  
# ===== LOOP =====  
while true; do  
    TS=$(date "+%Y-%m-%d %H:%M:%S")
```

```

# ----- TPS -----
STATS=$(mongostat -u "$MONGO_USER" -p "$MONGO_PASS" --authenticationDatabase "$AUTH_DB" --ssl --sslCAFile "$SSL_CA" --sslPEMKeyFile "$SSL_KEY")

INSERTS=$(echo "$STATS" | awk '{print $1}' | tr -cd "0-9")
UPDATES=$(echo "$STATS" | awk '{print $3}' | tr -cd "0-9")
DELETES=$(echo "$STATS" | awk '{print $4}' | tr -cd "0-9")

INSERTS=${INSERTS:-0}
UPDATES=${UPDATES:-0}
DELETES=${DELETES:-0}

# ----- CPU DELTA -----
curr_total=$(awk '/^cpu / {print $2+$3+$4+$5+$6+$7+$8}' /proc/stat)
curr_proc=$(awk -v pid=$MONGOD_PID '{print $14+$15}' /proc/$MONGOD_PID/stat)

total_diff=$((curr_total-prev_total))
proc_diff=$((curr_proc-prev_proc))

if [ "$total_diff" -gt 0 ] && [ "$proc_diff" -ge 0 ]; then
    CPU=$(awk "BEGIN {printf "%.2f", 100*$proc_diff/$total_diff}")
else
    CPU="0.00"
fi

prev_total=$curr_total
prev_proc=$curr_proc

# ----- REAL LATENCY -----
read curr_lat curr_ops <<< "$(
mongosh "mongodb://$MONGO_USER:$MONGO_PASS@$MONGO_HOST/$AUTH_DB" --tls --tlsCAFile "$SSL_CA" --sslPEMKeyFile "$SSL_KEY" --quiet --
s=db.getSiblingDB("admin").serverStatus().opLatencies;
lat=s.reads.latency+s.writes.latency+s.commands.latency;
ops=s.reads.ops+s.writes.ops+s.commands.ops;
print(lat);
print(ops);
' 2>/dev/null | tr -cd "0-9"
" | head -n 2
)"

curr_lat=${curr_lat:-0}
curr_ops=${curr_ops:-0}

delta_lat=$((curr_lat-prev_lat))
delta_ops=$((curr_ops-prev_ops))

# Clamp negatives
[ "$delta_lat" -lt 0 ] && delta_lat=0
[ "$delta_ops" -lt 0 ] && delta_ops=0

# Avg latency per operation in ms
if [ "$delta_ops" -gt 0 ]; then
    AVG_LATENCY=$(awk "BEGIN {printf "%.3f", $delta_lat/$delta_ops}")
else
    AVG_LATENCY="0.000"
fi

prev_lat=$curr_lat
prev_ops=$curr_ops

# CPU per operation

```

```

if [ "$delta_ops" -gt 0 ]; then
    CPU_PER_OP=$(awk "BEGIN {printf \"%.6f\", $CPU/$delta_ops}")
else
    CPU_PER_OP="0.000000"
fi

# ----- PROMETHEUS EXPORT -----
cat <<EOF > "$OUT_FILE"
# HELP mongo_cpu_percent MongoDB CPU usage percent
# TYPE mongo_cpu_percent gauge
mongo_cpu_percent $CPU

# HELP mongo_cpu_per_op MongoDB CPU per operation
# TYPE mongo_cpu_per_op gauge
mongo_cpu_per_op $CPU_PER_OP

# HELP mongo_avg_latency_ms MongoDB avg latency per operation (ms)
# TYPE mongo_avg_latency_ms gauge
mongo_avg_latency_ms $AVG_LATENCY

# HELP mongo_inserts_per_sec Mongo inserts per second
# TYPE mongo_inserts_per_sec gauge
mongo_inserts_per_sec $INSERTS

# HELP mongo_updates_per_sec Mongo updates per second
# TYPE mongo_updates_per_sec gauge
mongo_updates_per_sec $UPDATES

# HELP mongo_deletes_per_sec Mongo deletes per second
# TYPE mongo_deletes_per_sec gauge
mongo_deletes_per_sec $DELETES

# HELP mongo_total_ops_per_sec Mongo total operations per second
# TYPE mongo_total_ops_per_sec gauge
mongo_total_ops_per_sec $delta_ops
EOF

echo "$TS | Inserts/s: $INSERTS | Updates/s: $UPDATES | Deletes/s: $DELETES | TotalOps/s: $delta_ops | AvgLatency:"

sleep "$INTERVAL"
done

```

3. Make the Script Executable

```
sudo chmod +x /mongo/scripts/mongo_cpu_tps_latency_exporter.sh
```

Test-run it:

```
sudo /mongo/scripts/mongo_cpu_tps_latency_exporter.sh
```

4. Node Exporter Textfile Collector Setup

Create the directory for textfile metrics:

```

sudo mkdir -p /var/lib/node_exporter/textfile_collector
sudo chmod 755 /var/lib/node_exporter/textfile_collector

```

Start node_exporter with textfile collector enabled:

```
sudo pkill node_exporter
sudo /usr/local/bin/node_exporter \
  --web.listen-address=":9200" \
  --collector.textfile.directory="/var/lib/node_exporter/textfile_collector" &
```

Verify:

```
curl http://localhost:9200/metrics | grep textfile
curl http://localhost:9200/metrics | grep mongo_
```

5. Run the Exporter as a Systemd Service

Create unit file:

```
sudo nano /etc/systemd/system/mongo-metrics-exporter.service
```

Content:

```
[Unit]
Description=MongoDB TPS CPU Latency Exporter
After=network.target

[Service]
ExecStart=/mongo/scripts/mongo_cpu_tps_latency_exporter.sh
Restart=always
User=root

[Install]
WantedBy=multi-user.target
```

Enable and start:

```
sudo systemctl daemon-reload
sudo systemctl enable mongo-metrics-exporter
sudo systemctl start mongo-metrics-exporter
sudo systemctl status mongo-metrics-exporter
```

6. Prometheus Scrape Config

In prometheus.yml, ensure you scrape node_exporter:

```
scrape_configs:
  - job_name: "node_exporter"
    static_configs:
      - targets: ["mongodb-02:9200"]
```

Reload Prometheus after config change.

7. Grafana Dashboard Setup

In Grafana:

1. Add Prometheus as a data source (if not already).

2. Create a new dashboard.
3. Add the following panels:

Panel 1: "MongoDB CPU Usage (%)"

Query:
mongo_cpu_percent

Panel 2: "MongoDB Total Ops / Sec"

Query:
mongo_total_ops_per_sec

Panel 3: "MongoDB Inserts / Sec"

Query:
mongo_inserts_per_sec

Panel 4: "MongoDB Updates / Sec"

Query:
mongo_updates_per_sec

Panel 5: "MongoDB Deletes / Sec"

Query:
mongo_deletes_per_sec

Panel 6: "MongoDB Avg Latency (ms)"

Query:
mongo_avg_latency_ms

Panel 7: "MongoDB CPU Cost Per Operation"

Query:
mongo_cpu_per_op

You can set visualization as time-series and adjust units:

- CPU panels: Percent (0–100)
- Latency: milliseconds
- TPS / Ops: ops/sec

8. Basic Troubleshooting

1) If mongo_* metrics are missing in /metrics:

- Check that /var/lib/node_exporter/textfile_collector/mongo_ops.prom exists
- Ensure permissions are readable by node_exporter user

2) If latency is always 0:

- Check that `serverStatus().opLatencies` is enabled (MongoDB 3.4+)
- Ensure real load is running on the database

3) If CPU is 0:

- Check that mongod PID detection works:
pgrep -x mongod
- Confirm that mongod runs on this host

4) To simulate load:

- Run in mongosh:
use test
for (i=0;i<1000;i++) { db.t.insertOne({x:i}) }

Then re-check:

```
curl http://localhost:9200/metrics | grep mongo_
```

You should see non-zero TPS, latency, and CPU metrics.